

1. Model and likelihood

Let

$$\varepsilon_t = \Delta y_t = y_t - y_{t-1}, \quad t = 1, \dots, n,$$

with

$$\varepsilon_t \sim \text{i.i.d. Unif}(-b, b),$$

where the distribution is **symmetric around zero**, so $a = -b$ and $b > 0$ is the unknown parameter.

The density is

$$f(x; b) = \frac{1}{2b} \mathbf{1}_{\{|x| \leq b\}}.$$

The log-likelihood is therefore

$$\ell(b) = \sum_{t=1}^n \log f(\varepsilon_t; b) = -n \log(2b) + \sum_{t=1}^n \log \mathbf{1}_{\{|\varepsilon_t| \leq b\}}.$$

2. Simplification using the support constraint

Define

$$M = \max_{1 \leq t \leq n} |\varepsilon_t|.$$

Then

$$\sum_{t=1}^n \log \mathbf{1}_{\{|\varepsilon_t| \leq b\}} = \begin{cases} 0, & b \geq M, \\ -\infty, & b < M. \end{cases}$$

Hence,

$$\ell(b) = \begin{cases} -n \log(2b), & b \geq M, \\ -\infty, & b < M. \end{cases}$$

3. First derivative (score function)

For $b > M$,

$$\frac{\partial \ell(b)}{\partial b} = -\frac{n}{b} < 0.$$

For $b < M$, the log-likelihood is $-\infty$ and **not differentiable**.

✅ **Conclusion:**

The score is **strictly negative** on the admissible region (M, ∞) .

There is **no interior critical point**.

4. Second derivative

For $b > M$,

$$\frac{\partial^2 \ell(b)}{\partial b^2} = \frac{n}{b^2} > 0.$$

This confirms that $\ell(b)$ is **convex** for $b > M$ and therefore the maximum must occur on the **boundary of the parameter space**.

5. Maximum likelihood estimator

Since $\ell(b)$ is decreasing in b for $b \geq M$, the MLE is attained at the **smallest admissible value** of b , namely:

$$\hat{b} = \max_{1 \leq t \leq n} |\Delta y_t|$$

This is a **boundary MLE** (non-regular case).

6. Consequences for inference (important)

Because the MLE lies on the boundary:

- ❌ Fisher information is **not defined**
- ❌ Wald and score tests are **invalid**
- ✅ Exact and asymptotic inference must use **order-statistic theory**

7. Distribution of the MLE

Since $|\varepsilon_t| \sim \text{Unif}(0, b)$,

$$\frac{\hat{b}}{b} = \max_t \frac{|\varepsilon_t|}{b} \sim \text{Beta}(n, 1),$$

with CDF

$$\mathbb{P}(\hat{b} \leq x) = \left(\frac{x}{b}\right)^n, \quad 0 \leq x \leq b.$$

Hence,

$$\mathbb{E}[\hat{b}] = \frac{n}{n+1}b, \quad \text{Var}(\hat{b}) = \frac{n}{(n+1)^2(n+2)}b^2.$$

✔ Bias-corrected estimator:

$$\tilde{b} = \frac{n+1}{n}\hat{b}$$

8. Summary (ready to quote)

The log-likelihood corresponding to uniformly distributed increments with symmetric support $[-b, b]$ is not differentiable in the usual sense. The score function is strictly negative on the admissible region, implying the absence of an interior maximum. The maximum likelihood estimator of b is therefore attained on the boundary of the parameter space and is given by the maximum absolute increment.

$$\hat{b} = \max_t |y_t - y_{t-1}|.$$

Model recap

Let

$$\varepsilon_t = y_t - y_{t-1} \sim \text{i.i.d. Unif}(-b, b), \quad t = 1, \dots, n.$$

Define

$$\hat{b} = \max_{t=1, \dots, n} |\varepsilon_t|.$$

1. Distribution of the MLE

Since $|\varepsilon_t| \sim \text{Unif}(0, b)$,

$$\frac{\hat{b}}{b} = \max_{1 \leq t \leq n} \frac{|\varepsilon_t|}{b} \sim \text{Beta}(n, 1).$$

Therefore, for $0 \leq x \leq b$,

$$\mathbb{P}(\hat{b} \leq x) = \left(\frac{x}{b}\right)^n.$$

2. Variance of the estimator

Mean

$$\mathbb{E}[\hat{b}] = \frac{n}{n+1}b \quad (\text{downward biased}).$$

Variance

$$\text{Var}(\hat{b}) = \frac{n}{(n+1)^2(n+2)}b^2$$

Bias-corrected estimator

$$\tilde{b} = \frac{n+1}{n}\hat{b}, \quad \text{Var}(\tilde{b}) = \frac{b^2}{n(n+2)}.$$

3. Exact $(1 - \alpha)$ confidence interval for b

From

$$\mathbb{P}(\hat{b} \leq bu) = u^n,$$

we obtain:

we obtain:

$$\mathbb{P}(\hat{b} \geq b \alpha^{1/n}) = 1 - \alpha.$$

✓ Exact one-sided lower CI

$$\left[b \geq \frac{\hat{b}}{\alpha^{1/n}} \right]$$

✓ Exact central two-sided CI

$$\left[\hat{b}, \frac{\hat{b}}{(\alpha/2)^{1/n}} \right]$$

Note: the lower bound cannot be smaller than $\hat{b}$ (boundary estimator).

6. Summary (ready to quote)

$$\hat{b} = \max_t |y_t - y_{t-1}|, \quad \text{Var}(\hat{b}) = \frac{n}{(n+1)^2(n+2)} b^2.$$

$$\text{CI}_{1-\alpha}(b) = \left[\hat{b}, \hat{b}(\alpha/2)^{-1/n} \right].$$